Appendix A

Method for matching representation cases to ULP charges

A.1 Structure of the records

The NLRB's fundamental unit of record-keeping is the case. If you file a petition to hold an election, the NLRB opens a new representation case record (I will call these "R-cases"). If you file a ULP charge, the NLRB opens a new complaint case record ("C-cases"). The complication of this record-keeping structure is that the unit of interest in this study is the union organizing drive. Every drive has at least one R-case record and may have one or more C-case records. Looking at the NLRB's case records, though, is like pulling all of the receipts from the dumpster outside of a large apartment building. You cannot make inferences about the residents' shopping patterns until you figure out which receipts belong to which residents. Similarly, we cannot make inferences about ULP charges in organizing drives until we figure out which C-cases go to which R-cases, if any. We need a way to match the two types of cases.

In the past and after 2007, the AFL-CIO relied on the employer's address to do this matching. When the NLRB debuted out its new database, CATS, in 1999, however, it decided that employer's address would no longer be discoverable, even through a Freedom of Information Act request.

A.1.1 "Backing out" ULPs: an outer join

Since I lacked a unique matching identifier for these data, I had to perform an outer join on the data. An outer join involves taking some *group* identifier present in each dataset and creating all possible combinations of records that share the group identifier, i.e., the Cartesian product of the datasets.

Picture that we have two datasets, one of R-cases and one of C-cases. Each dataset has eight records. Each record has a unique identifier for the record and a group ID that may or may not be unique. That group ID may appear in one or both of the datasets. To be more specific, assume that the R-case dataset looks like this:

R-case	GroupID
0	2
1	3
2	3
3	4
4	5
5	5
6	5
7	6

And assume the C-case dataset looks like this:

C-case	GroupID
1	6
2	5
3	5
4	5
5	4
6	3
7	3
8	1

If we were to perform an outer join on these two datasets based on the group ID, the resulting dataset would look like this:

R-case	GroupID	C-case
.	1	8
0	2	.
1	3	7
1	3	6
2	3	7
2	3	6
3	4	5
4	5	4
4	5	3
4	5	2
5	5	4
5	5	3
5	5	2
6	5	4
6	5	3
6	5	2
7	6	1

Some things should be noted about this procedure. First, one of the C-cases (8) had a group ID that was not present the R-case data; thus it has no matches at all. Similarly, one R-case (0) has no matches in the C-case data. Second, Two of the R-cases (3 and 7) had group IDs that uniquely matched records in the C-cases (5 and 1, respectively). In these cases, the outer join will produce one record. Third, in those cases where n R-cases and n C-cases had the same group ID, the outer join produced $n \times n$ records—every pairwise combination possible.

I created a group ID for each R-case and each C-case based on the Metropolitan Statistical Area (MSA) and six-digit industry (per the 2002 NAICS codes) noted for each record. I then performed an outer join based on that group ID. In other words, I formed every pairwise combination of R-cases and C-cases in a given six-digit industry in a given city during the period. My goal was to "back out" the organizing drives that had ULP charges. I would create all those combinations and then strike out the combinations that did not make sense, based on criteria I describe below. The ones that remained after the strikeouts would be the ones I would assume had ULP charges.

A.1.2 Firm versus soft matches

Before I started striking out cases, I first classified most of the cases as firm yesses and noes, where "yes" and "no" refer to whether a given R-case had a ULP charge associated with it.

Many of the R-cases were like (0) in my example above—they had no matching C-cases in their city and industry. I classified these R-cases as firm noes. Of the 22,382 records that I began the analysis with, 12,151 (54.3 percent) were firm noes.

It would be tempting to assume that R-cases like (3) and (7) above, those with single C-case matches, were firm yesses. This is wrong, though. That single C-case could have been filed by an existing unit in the same city and industry. I needed another method for looking for firm yesses. I looked for cases where there was *documentary evidence* of matching; i.e., where either the R- or the C-case mentioned the other explicitly. I looked for six things:

1. If the R-case mentions one of the following:

   (a) The ID of a C-case, which is the same as the ID of the C-case shown as matched.

   (b) The ID of a C-case cited in a blocking order, which is the same as the ID of the C-case shown as matched.

   (c) The ID of a C-case cited in a request to proceed, which is the same as the ID of the C-case shown as matched.

(d) The ID of a C-case cited in a request for a 10(j) injunction, which is the same as the ID of the C-case shown as matched.

2. If the C-case mentions one of the following:

   (a) The ID of a R-case, which was the same as the ID of the R-case shown as matched.

   (b) The name and unit code of the case (i.e., of the employer and the NLRB's tentative bargaining-unit designation), which was the same as the name and unit code of the R-case shown as matched.

Under these six conditions, where I had documentary evidence of a specific representation case citing a specific complaint case or vice versa, I assumed that the match was a firm yes. Of the 22,382 cases that I began the analysis with, 2,881 (12.9 percent) were firm yesses.

Marking 15,032 records as firm yesses and noes took care of 67.2 percent of the records. There remained 7,350 R-cases that had one or more C-cases in the same city and industry but that lacked documentary proof that they were matched. To these I now turned.

A.1.3 Backing out soft matches

I used many dates to evaluate matches. This is because they were widely available. The NLRB's CATS database either automatically enters dates for events (such as the opening or closing of a case) or requires the user to input them (such as the date of an election).[1] Some combinations of dates were extremely unlikely to happen in the cases I was looking for. For example, if a C-case was closed before the R-case was filed, then even though the two cases were in the same city and industry, that ULP charge probably was not associated with that organizing drive. I looked for the following things:[2]

[1]These user-entered fields can still be blank, because the user can note that a date is at present unknown.

[2]For several of these, note that I was looking for *pre-election* ULP charges. Subsequent charges, as for example relating to the conduct of the election, are picked up in the later-stage regression

1. The closing date for the R-case is before the filing date of the C-case.

2. The election date for the R-case is before the filing date of the C-case.

3. The date a complaint (the NLRB's determination of fault) is mentioned being issued in the R-case is before the filing date of the C-case.

4. The date a blocking order was issued in the C-case is before the filing date of the R-case.

5. The date a complaint was issued in the C-case is before the filing date of the R-case.

6. The date an unblocking order was issued in the C-case is before the filing date of the R-case.

7. The date that of a hearing or decision by the regional director or the Board occurred is before the filing date of the R-case.

My reasoning was that if the R-case and C-case did not overlap in time, then the one was probably not associated with the other. These seven rules together struck out all of the potentially matching C-cases for 4,932 of the 7,350 undetermined R-cases. Continuing with the metaphor, I will call these records soft noes. This left 2,418 R-cases undetermined.

Next I checked the type of ULP charge mentioned in the C-case. Some types of ULP charges can only be filed by established bargaining units and thus would not count as matches here. Ruling out such matches moved 428 additional undetermined cases to the soft noes column, leaving 1,990.

I then lexicographically sorted the remaining undetermined cases. Such a sort is very useful for finding duplicated records. I looked for two types of duplicates: trivial, defined as two records with the same R-case or C-case ID where the only difference is that one record has information in a field left blank in the other; and non-trivial, where otherwise identical records conflicted in one field—e.g., one had the election date as 08/15/02 and the other had the election date as 08/21/02. I assumed that

pertaining to reaching a first contract.

the 567 trivial duplicates were trivial[3] and combed them out. I then resolved the 95 non-trivial duplicates by hand.

All told, I coded 5,594 of the 7,350 undetermined matches as soft noes. This left 1,756 potential R-case/C-case matches that happened in the same city, in the same industry, at the same time, in units with similar numbers of workers, with the "right" type of ULP charge and with no other discrepancies that I could find in the available information. I coded these as soft yesses.

A.2 Evaluating the methodology

I dislike using words like "firm" and "soft" here. Those so-called soft matches are the best I, or I think anyone, could do with the available data. I am nonetheless drawing the distinction because I can gain some leverage from it to test what the potential sources of bias in this matching method may be. Specifically, I can contrast summary statistics for the firm and soft matches to see how they differ from one another, and I can compare regression results for just the firm matches with results for the full sample. I discuss both of these tests below.

A.2.1 Distribution of firm and soft matches

Table A.1 shows the 2×2 breakdown of whether I assumed a case had a ULP charge associated with it and whether it was a firm yes or no. The bulk of the records, and the bulk of both the yesses and the noes, were firm assignments. In table A.1, the upper-right numbers in each cell show the cell's within-row marginal probability (i.e., summing them across the row yields one) and the lower-left numbers show the cell's within-column marginal probability (i.e., summing them down the column yields

[3]As mentioned in the paper, these duplicates could be created by the database itself. Sometimes when the user updated a record the database would save the changes as a new record rather than overwrite the old record.

Table A.1: Breakdown of ULP charges, by firm and soft matching

	Firm match	Soft match	Total
ULP	(.621) 2881 (.192)	(.379) 1756 (.239)	4637 (.207)
No ULP	(.685) 12151 (.808)	(.315) 5594 (.761)	17745 (.793)
Total	(.672) 15032	(.328) 7350	22382

Cells show frequency counts. Row marginal probabilities are given in the upper right of each cell, and column marginal probabilities are given in the lower left.

one).

A quirk of this table is that the share of firm and soft cases ruled as yesses is quite similar—about a fifth in each case. This means nothing.

Subsequent research (forthcoming) using slightly different samples of NLRB case data have found higher rates of ULP charges associated with representation cases—in some instances substantially higher. This begs the question: whence the discrepancy? I will mention three possiblities; there could be more. First, I matched these cases based on comparing date fields and similar information in the two cases. If one of the two records lacked one of the date fields, then I could mae no such comparison. In those cases I erred toward assuming that the case did *not* have a ULP charge. Second, if organizers filed a ULP charge during the card drive and then filed an election petition while the C-case was still open, my procedure would almost always code that case as a no. If I could link the employer more easily, then I could investigate whether some such cases should instead be yesses. Third, any errors (or empty fields) in the MSA or 6-digit industry code would affect my initial outer join.

In these and other scenarios, I would have coded some records that really had ULP charges as not having them. The fundamental point is that, because I was backing out matches from imperfect data, I wanted to be extremely conservative in my assumptions. This was an intellectual and a political decision. Intellectually, I wanted to avoid sources of bias that would overstate my findings. Politically, I wanted if anything to undercount cases that had ULP charges, lest I be accused of cherry-picking non-ULP cases so as to make the effects of ULPs seem larger than they are.

A.2.2 Measurement-error bias

What are the implications of this approach for my results? Since I have some firm yesses and noes, I can see how they differ from the soft yesses and noes. I ran a battery of t-tests and χ^2-tests to compare the firm and soft yesses and the firm and soft noes and to contrast the firm yesses and noes with the soft yesses and noes. That is to say, I tested for differences between the cells of the 2×2 table shown above. For the sake of space, I will summarize those test results by saying that the differences between the soft yesses and noes are smaller than between the firm yesses and noes, but that the soft yesses still look more like the firm yesses than like the soft noes. In other words, there is some measurement error in assignment, but not a crippling amount.

Such measurement error should bias the estimated coefficients in the regression results toward zero. In other words, the "real" coefficients on ULP charges should be *even larger*—even more negative, in this instance. To see whether this was so, I replicated the paper's regressions on the subsample of records that were firm yesses or noes. Table A.2 shows those results (I have ommitted other coefficients for space). As expected, considering only the firmly classified cases produces even larger coefficients that remain significant. In fact, the coefficient of 8(a)(3) charges in holding an election, which was not significant in the full sample, is significant in the limited sample.

Table A.2: Test for measurement-error bias: comparison of estimated ULP-charge coefficients using full and limited samples

	Election held		Election won		Contract reached	
	Full	Limited	Full	Limited	Full	Limited
8(a)(1) charge filed	-.54	-.74	-.34	-.34		
8(a)(3) charge filed	*-.13*	-.28	*.03*	*-.17*		
Other charge filed	-.71	-.87	*-.04*	*-.14*		
ULP during negotiations					-1.66	-1.90
8(a)(1) before election					-.44	-.75
8(a)(3) before election					*-.22*	*-.30*
Other before election					*.27*	*.15*
8(a)(1) before, ULP during					-1.91	-2.41
8(a)(3) before, ULP during					-1.94	-2.47
Other before, ULP during					-1.48	-1.93
N	14424	8855	9593	5885	2499	1427

The coefficients reported for the full sample are drawn from an earlier draft of chapter 2 of this thesis: John-Paul Ferguson, "The Eyes of the Needles: A Sequential Model of Union Organizing Drives" (MIT-IWER Working Paper, February 2007). The coefficients for the limited sample were produced by rerunning the model on a sub-sample consisting of cases with a firm R-case/C-case match, as described herein. Coefficients in italics were not significant at the .05 level.

A.3 Implications

I close with some thoughts on how this study would relate to others. This study estimates a lower bound on the effect of ULP charges. At each point where I had to make a judgment call about whether to match an organizing drive with a ULP charge, I chose *not* to match them unless there was direct evidence in favor of matching. I chipped away at the undetermined R-cases, trying to reduce the number of cases that I would say had ULP charges just based on my inference.

At each point in that process, I had in mind the potential critic who would accuse me of assigning cases so as to play up the damaging effect of ULP charges. As of spring 2009, it so happens that I am now receiving questions about whether I have played *down* their effect. In a sense, this is a relief.

How do we treat a lower bound? When someone asks us what the effect of a ULP charge is on an organizing drive's chances of success, we say "we estimate that, on average, a ULP charge is associated with *at least* a 30-percent reduction in the likelihood..." etc. In the final analysis, then, this appendix boils down to two words: "at least."